

BIKANER, India

WMO: 421650

Lat: 28.0207N

Lon: 73.2772E

Elev: 735

StdP: 14.31

Time Zone: 5.50 (IND)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 43.1	(c) 45.7	(d) 16.9	(e) 13.2	(f) 72.0	(g) 21.6	(h) 16.6	(i) 71.5	(j) 10.3	(k) 65.2	(l) 7.7	(m) 63.2	(n) 0.9	(o) 50	(p) 0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 19.0	(c) 111.4	(d) 70.8	(e) 109.1	(f) 71.6	(g) 106.9	(h) 72.3	(i) 82.6	(j) 94.0	(k) 81.6	(l) 93.1	(m) 80.8	(n) 92.4	(o) 4.7	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 80.1	(b) 160.6	(c) 87.0	(d) 79.1	(e) 155.3	(f) 86.6	(g) 78.2	(h) 150.7	(i) 86.6	(j) 47.4	(k) 93.8	(l) 46.3	(m) 92.9	(n) 45.4	(o) 92.3	(p) 88.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	11.7	9.2	37.3	114.6	3.0	2.3	35.1	116.3	33.4	117.6	31.7	119.0	29.5	120.7
			33.7	84.6	2.4	1.6	32.0	85.8	30.6	86.7	29.3	87.6	27.6	88.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.3	59.8	67.5	78.2	87.9	95.2	95.9	92.3	89.7	88.3	83.5	73.4	63.2
	DBStd	13.02	5.17	5.83	6.59	5.45	4.76	4.26	3.96	3.10	3.18	4.60	5.02	5.34
	HDD50	2	2	0	0	0	0	0	0	0	0	0	0	0
	HDD65	317	175	36	2	0	0	0	0	0	0	0	5	99
	CDD50	11429	307	489	873	1138	1401	1376	1313	1230	1150	1039	703	410
	CDD65	6268	14	106	410	688	936	926	848	765	700	574	258	43
	CDH74	92089	153	1018	4765	10256	15830	15653	13254	11288	9952	7083	2420	417
	CDH80	58742	12	319	2501	6572	11457	11370	8821	6849	5813	3959	996	73
Wind	WSAvg	2.6	1.5	2.1	2.4	2.8	3.9	4.7	4.1	3.2	2.5	1.7	1.2	1.1
Precipitation	PrecAvg	11.7	0.2	0.4	0.3	0.2	0.9	1.6	3.8	2.4	1.3	0.4	0.1	0.1
	PrecMax	24.5	1.1	4.1	2.4	2.0	3.7	8.0	14.6	8.0	7.8	5.0	1.7	1.7
	PrecMin	2.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	4.8	0.2	0.7	0.6	0.5	1.1	1.6	3.1	1.7	1.6	0.9	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.8	92.7	102.6	109.7	114.9	114.2	108.8	104.5	104.8	102.5	93.6	85.5
		MCWB	58.2	62.1	64.2	67.6	69.7	71.4	78.3	79.0	73.2	67.8	64.3	61.6
	2%	DB	78.3	87.4	99.1	106.9	112.4	110.6	105.8	101.5	102.1	99.8	90.9	82.2
		MCWB	56.7	61.4	63.9	66.9	69.9	73.1	78.0	78.7	73.2	69.4	64.1	60.2
	5%	DB	75.2	84.0	95.9	104.4	110.2	108.4	103.4	99.6	99.8	97.6	88.7	79.1
		MCWB	55.7	60.5	63.4	66.6	69.8	74.3	78.3	78.3	74.4	69.0	63.6	58.5
	10%	DB	72.4	80.5	92.4	101.7	107.7	106.1	101.3	97.7	97.6	95.6	86.2	75.7
		MCWB	54.4	58.8	62.4	66.2	70.3	74.8	78.2	78.2	74.9	68.4	62.9	56.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.7	69.2	70.6	75.2	80.3	83.3	83.8	83.9	83.4	78.0	71.1	65.2
		MCDB	73.8	80.9	88.8	100.3	100.6	99.9	95.8	93.5	93.9	91.6	85.7	79.8
	2%	WB	59.7	65.6	67.8	71.5	78.0	81.0	82.4	82.4	81.5	75.7	67.9	62.7
		MCDB	71.9	78.9	85.7	95.7	97.8	96.9	93.6	93.1	92.2	89.5	84.7	76.6
	5%	WB	57.7	63.3	66.2	69.7	76.5	79.7	81.6	81.4	80.3	73.9	65.6	60.6
		MCDB	70.2	77.5	85.9	95.0	96.7	96.7	92.9	91.8	90.7	88.5	84.2	73.6
	10%	WB	55.9	61.3	64.8	68.1	75.3	78.9	80.8	80.6	79.2	71.7	63.9	58.5
		MCDB	69.0	75.0	84.6	94.3	96.3	96.6	92.4	90.8	88.9	88.9	82.7	73.2
Mean Daily Temperature Range	5% DB	MDBR	23.2	23.1	23.8	24.0	22.9	19.0	16.0	15.1	17.7	23.0	25.3	24.7
		MCDBR	26.0	26.2	26.4	25.9	24.9	21.1	18.2	17.3	20.4	24.3	26.5	27.2
		MCWBR	12.3	10.6	8.9	9.0	8.2	6.5	4.2	4.3	6.0	8.2	11.3	12.5
	5% WB	MCDBR	22.3	21.9	22.7	22.9	21.6	18.0	16.5	14.9	15.8	20.9	24.2	23.7
		MCWBR	11.8	10.1	8.2	9.4	9.5	5.8	4.6	4.8	5.5	9.4	11.3	11.7
Clear-Sky Solar Irradiance	taub		0.432	0.438	0.479	0.561	0.643	0.716	0.740	0.641	0.548	0.540	0.502	0.457
	taud		1.970	1.997	1.900	1.741	1.614	1.539	1.531	1.719	1.876	1.780	1.816	1.906
	Ebn at Noon		248	260	258	241	221	205	199	219	236	226	223	232
	Edh at Noon		49	52	61	74	84	90	90	75	62	64	57	51
All-Sky Solar Radiation	RadAvg		1217	1535	1912	2164	2249	2087	1898	1850	1847	1669	1315	1165
	RadStd		64	64	69	60	49	95	95	115	89	65	70	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page